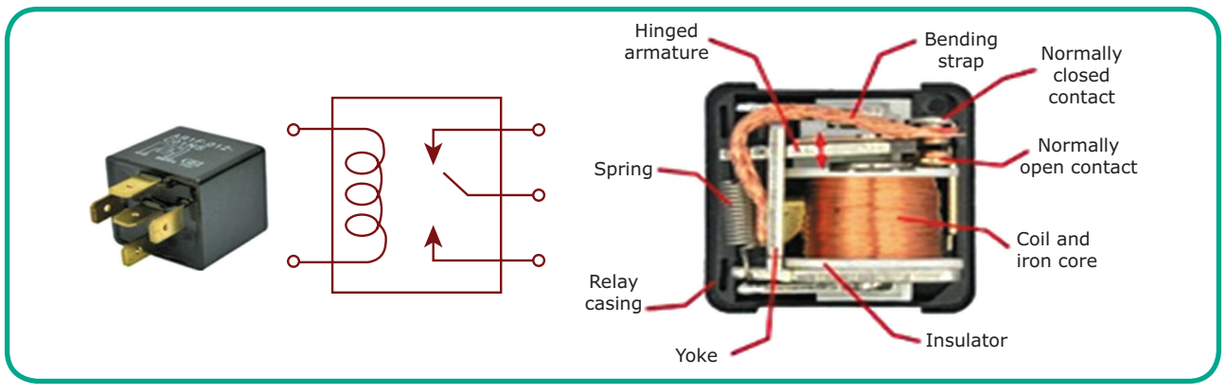


Figure 2.13 Relay

2.2.7 Action of Coils in AC and DC Circuits

2.2.7.1 AC Circuits

Since the phase is alternatively changed in AC circuits, the back EMF continuously presents in the coil. This back EMF depends upon the frequency and the core of the coil. Hence, the coils in AC circuits have resistance carried by back EMF along with normal resistance. These two resistances combined together is known as inductive reactance.

$$\text{Inductive Reactance } X_L = 2\pi fL$$

where $\pi = 3.14$, f is the frequency of AC and L is the inductance. In AC circuits, the current flows in the coil lag 90° with reference to the voltage.

2.2.7.2 DC Circuits

When the switch is ON, the back EMF generated opposes the flow of current. The back EMF will get vanished when constant magnetic field is produced, i.e. the back EMF persists only to the fraction of switching on the circuit. When the circuit is switched ON, automatically the back EMF becomes zero. In DC circuits, coils would have only normal resistance. Hence, it may not be taken into consideration.

2.3. TRANSFORMERS

A device which has two or more coils wound on a core is termed as transformer. The electrical energy is transformed from one coil to other through mutual inductance in it.

The symbol of Transformer is shown in Figure 2.14

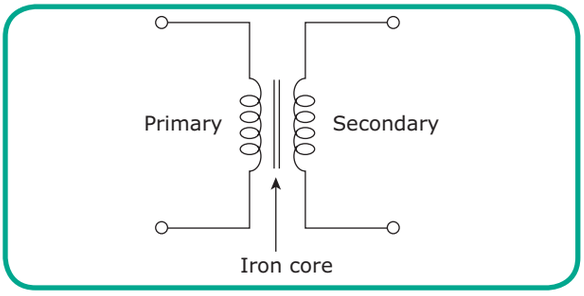


Figure 2.14 Symbol of Transformer

History

In 1880s, the first commercially-used transformer was built by William Stanley, working under George Westinghouse that was wound to form a core of E-shaped plates in step-up and step-down variations.

2.3.1 Principle of Transformer Construction

Transformer is a static device which contains one primary and one or more secondary coils. The electrical energy in

ELEC_EQUIP_Chapter02.indd 44 11/10/2020 12:23:49 PM